MILITARY SPECIFICATION

FLARE, SURFACE, TRIP, M49A1
PARTS, AND LOADING, ASSEMBLING AND PACKING

This specification is approved for use by U.S. Army Armament Research and Development Command and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of parts for the loading, assembling and packing for one type of flare designated as Flare, Surface, Trip, M49A1 assembly (Alt) and Flare, Surface, Trip, M49A1 assembly (Encapsulated binder).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416 - Plating, Cadmium (Electrodeposited)

MILITARY

MIL-P-20444 - Primer, Percussion, M42, Loading, Assembling and Packing
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

FSC: 1370

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:46Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 -	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-331 -	Fuze and Fuze Components, Environmental and Performance Tests For
MIL-STD-1234-	Pyrotechnics: Sampling, Inspection and Testing

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.4)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT AND PACKAGING DRAWINGS

8836957 -	Flare, Surface, Trip, M49A1, Assembly
9288866 -	Flare, Surface, Trip, M49A1, Assembly (Alt)
9313553 -	Flare, Surface, Trip, M49A1, Assembly (Encapsulated Binder)
8830880 -	Box, Packing, Ammunition, for Flare, Surface, Trip, M49A1
8830881 -	Box, Packing, Ammunition, for Flare, Surface, Trip, M49A1
9277238 -	Box, Fiberboard, Packing, Ammunition, for Flare, Surface, Trip, M49A1 (Alt)
9277239 -	Box, Packing, Ammunition, for Flare, Surface, Trip, M49A1 (Alt)

INSPECTION EQUIPMENT

9201136 -	Tunnel, Light
9201268 -	Procedure, Light Output Measurement
9201390 -	Procedure, Photocell Checkout
9247118 -	Pull Pin Test Fixture
9247119 -	Operating Instructions for Pull Pin Test Fixture

PUBLICATIONS

Analytical Methods for Atomic Absorption Spectrophotometer - Perkins - Elmer, Revision March 1971

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings, specifications and standards.

3.2 Components and assemblies. The components and flare assemblies shall comply with all requirements specified on Drawing. (dwg.) 8836957, 9288866 and 9313553 when applicable, and with all requirements specified in applicable specifications and standards.

3.3 Embrittlement of springs and retainer. The striker spring (see dwg. 8864755), the trigger spring (see dwg. 8836975) and the retainer (see dwg. 9282921), shall not break when stretched to the minimum (min.) point of permanent distortion.

3.4 Jumble test. The flare assembly shall comply with the following requirements.

3.4.1 Flare assembly. The flare assembly shall not function during the test.

3.4.2 Safety clip. The safety clip assembly shall not be totally or partially disengaged after test.

3.4.3 Flare assembly. The flare assembly shall be safe to transport after test (see 6.6).

3.4.4 There shall be no evidence of external damage to the flare assembly that will affect the intended function (see 6.7).

3.5 Functioning. The flare shall ignite and shall comply with the following requirements.

3.5.1 Ambient temperature (21±2°C) (70±5°F). The flare assembly shall ignite without exploding, and burn for the required time when a pull force of not less than 1.50 nor more than 7.00 pounds is applied to the trip wire when attached to the pull pin and a pull force of not less than 2.00 nor more than 9.00 pounds is applied to the trip wire when attached to the trigger.

3.5.2 Cold temperature (minus 34°±2°C) (minus 30±5°F). The flare assembly shall comply with the requirements of 3.5.1.

3.5.3 Temperature and humidity cycling. The flare assembly shall comply with the requirements of 3.5.1 except that the pull pin force shall be not less than 1.50 nor more than 8.00 pounds.

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:46Z
Check the source to verify that this is the current version before use.

3.6 Water submersion test. The packed carton sealed in the barrier bag shall show no evidence of water leakage.

3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection .

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the items in] sample quantities as indicated in Table I.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
	Flare, Surface, Trip, M49A1				See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Case (Prior to coating) (Dwg. 8836968 or 9288872) Examination of defects	25	-	3.2	4.4.2.1
	Lever (Dwg. 8836961) Examination of defects	25	-	3.2	4.4.2.2
	Striker (Dwg. 8836972) Examination of defects	25	-	3.2	4.4.2.3
	Striker Assembly (Dwg. 8836971) Examination of defects	25	-	3.2	4.4.2.4
	Hinge Pin (Dwg. 8836962) Examination of defects	25	-	3.2	4.4.2.5
	Mounting Bracket (Dwg. 8836978) Examination of defects	25	-	3.2	4.4.2.6
	Trigger (Dwg. 8836976) Examination of defects	25	-	3.2	4.4.2.7

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
	Flare, Surface, Trip, M49A1				See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Trigger Assembly (Dwg. 8836973)				
	Examination of defects	25	-	3.2	4.4.2.8
	Trigger Pivot (Dwg. 8836974)				
	Examination of defects	25	-	3.2	4.4.2.9
	Pull Pin (Dwg. 8836960)				
	Examination of defects	25	-	3.2	4.4.2.10
	Spring, Striker (Dwg. 8864755)				
	Examination of defects	25	-	3.2	4.4.2.11
	Test for spring embrittlement	10	-	3.3	4.5.1
	Safety Clip (Dwg. 8836958)				
	Examination of defects	25	-	3.2	4.4.2.12
	Cover (Dwg. 8836965)				
	Examination of defects	25	-	3.2	4.4.2.13

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-01-19T07:40Z
Check the source to verify that this is the current version before use.

TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
	Flare, Surface, Trip, M49A1				See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Spring Trigger (Dwg. 8836975)				
	Examination of Defects	25	-	3.2	4.4.2.14
	Test for Spring Embrittlement	10	-	3.3.	4.5.1
	Spool Retainer (Dwg. 9282921)				
	Examination of defects	25	-	3.2	4.4.2.15
	Spring test	10	-	3.2	4.5.12
	Wire Spool Assembly (Dwg. 9269197)				
	Examination of defects	25	-	3.2	4.4.2.16
	Cover Loading Assembly (Dwg. 8836964)				
	Examination of defects	25	-	3.2	4.4.2.17
	Flare Assembly (Dwg. 8836957, 9313553 or 9288866)		-		
	Examination of defects	148	-	3.2	4.4.2.19
	Jumble test	25	-	3.4	4.5.2
	Functioning (ambient temp.)	28	-	3.5.1	4.5.4.1
	Functioning (cold temp.)	25	-	3.5.2	4.5.4.2
	Functioning (Temperature and humidity cycling)	50	-	3.5.3	4.5.4.3
	Candlepower test	20	-	3.2	4.5.6

NOTES:

DRDAE-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 20260110-191074602
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of trip flares shall contain:

a. Parts of one interfix lot number from one supplier.
b. Primers of one interfix lot number from one manufacturer.
c. Composition ingredients from not more than one lot.
d. Illuminant composition produced by one manufacturer under one continuous set of operating conditions. and which consists of one or more batches that have been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.

4.4.2 Examination. (See MIL-A-48078). Unless otherwise specified in the classification of defects an test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Case, Prior to Coating		8836968 or 9288872
			NEXT HIGHER ASSEMBLY 8836957, 9313553 or 9288866

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wall thickness, min.		0.40%	3.2	Gage
102	Thickness of bottom excluding angular surface		0.40%	3.2	Gage
103	Inside diameter at mouth		0.40%	3.2	Gage
104	Crack, fold, inclusion or similar imperfection		0.40%	3.2	Visual
105	Radius at bottom outside, excessive		0.40%	3.2	Visual
Minor					
201	Inside diameter below mouth area, min.		0.65%	3.2	Gage
202	Inside diameter, maximum (max.)		0.65%	3.2	Gage
203	Total length		0.65%	3.2	Gage
204	Outside diameter, max.		0.65%	3.2	Gage
205	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-01-19T07:46Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.2	Lever				8836961
					NEXT HIGHER ASSEMBLY 8836957, 9288866 or 9313553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness of material, min.		0.40%	3.2	Gage
102	Length from hinge surface to underside of top surface		0.40%	3.2	Gage
103	Length from hinge to hinge, min.		0.40%	3.2	Gage
104	Width of hinge, min.		0.40%	3.2	Gage
105	Radius of hinge		0.40%	3.2	Gage
106	Length from inside surface of hinge to tip, min.		0.40%	3.2	Gage
107	Crack		0.40%	3.2	Visual
108	Height of bend end		0.40%	3.2	Gage
109	Length of turn-up at end of handle, min.		0.40%	3.2	Gage
Minor					
201	Length between sides of top surface, max.		0.65%	3.2	Gage
202	Length from centerline of hinge to curve of handle (includes location and radius)		0.65%	3.2	Gage
203	Length from centerline of hinge to handle, max.		0.65%	3.2	Gage
204	Width of large dimple on top surface		0.65%	3.2	Gage
205	Height of large dimple on top surface		0.65%	3.2	Gage
206	Length of large dimple on top surface		0.65%	3.2	Gage

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.2	Lever				8836961
					NEXT HIGHER ASSEMBLY 8836957, 9288866 or 9313553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
207	Total length (without dimple)		0.65%	3.2	Gage
208	Width of tail of large dimple on top surface		0.65%	3.2	Gage
209	Length from end of tail of large dimple to edge of handle		0.65%	3.2	Gage
210	Height of tail of large dimple		0.65%	3.2	Gage
211	Small angle of inclined dimpled surface		0.65%	3.2	Gage
212	Improper finish		1.00%	3.8	Visual
213	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current revision before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Striker				8836972
					NEXT HIGHER ASSEMBLY 8836957, 9288866 or 9313553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length over hinge pin tabs		0.40%	3.2	Gage
102	Thickness, min.		0.40%	3.2	Gage
103	Location of firing pin hole in relation to hinge pin tabs and holes		0.40%	3.2	Gage
104	Diameter of hinge pin holes		0.40%	3.2	Gage
Minor					
201	Diameter of hinge pin tabs, min.		0.65%	3.2	Gage
202	Diameter of firing pin hole, max.		0.65%	3.2	Gage
203	Angle of firing pin hole surface		0.65%	3.2	Gage
204	Bare spot in protective finish		0.65%	3.8	Visual
205	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Striker Assembly		8836971
			NEXT HIGHER ASSEMBLY 8836957, 9288866 or 9313553

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length from point to pivot hole		0.40%	3.2	Gage
102	Length over hinge pin tabs		0.40%	3.2	Gage
103	Profile of point		0.40%	3.2	Gage
104	Striker point loose		0.40%	3.2	Manual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Hinge Pin				8836962
					NEXT HIGHER ASSEMBLY 8836957, 9288866 or 9313553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length, min.		0.40%	3.2	Gage
102	Diameter, max.		0.40%	3.2	Gage
Minor					
201	Diameter, min.		0.65%	3.2	Gage
202	Finish improper		0.65%	3.8	Visual
203	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-01-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Mounting Bracket (Prior to Painting)		8836978
			NEXT HIGHER ASSEMBLY 8836957, 9288866 or 9313553

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Location of slot		0.65%	3.2	Gage
202	Length from top of lugs to pointed end, min.		0.65%	3.2	Gage
203	True position of carriage bolt holes		0.65%	3.2	Gage
204	True position of trigger pivot hole, min.		0.65%	3.2	Gage
205	Crack		0.65%	3.2	Visual
206	Any operation missing or incorrect		0.65%	3.2	Visual
207	Width of mounting surface		0.65%	3.2	Gage
208	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil - Downloaded: 2026-01-19T07:46Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Trigger		8836976
			NEXT HIGHER ASSEMBLY 8836957; 9288866 or 9313553

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Width of spring lug, min.		0.40%	3.2	Gage.
102	Thickness of material, min.		0.40%	3.2	Gage
103	Width of lever holding finger, min.		0.40%	3.2	Gage
104	Crack		0.40%	3.2	Visual
105	Inside height of lever holding finger		0.40%	3.2	Gage
Minor					
201	Length from top to base surface, min.		0.65%	3.2	Gage
202	Length from pivot hole to shoulder slot		0.65%	3.2	Gage
203	Length from pivot hole to inside of spring lug		0.65%	3.2	Gage
204	Diameter of pivot hole		0.65%	3.2	Gage
205	Any operation missing or incorrect		0.65%	2.2	Visual
206	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

Source: https://assistdla.mil -- Downloaded: 2026-01-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Trigger Assembly				8836973
					NEXT HIGHER ASSEMBLY 8836957, 9288866 or 9313553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Trigger pivot loose		0.40%	3.2	Manual
102	Trigger bound		0.40%	3.2	Manual
103	Any component missing or broken		0.40%	3.2	Visual
104	Component incorrectly assembled		0.40%	3.2	Visual
105	Trigger spring insecurely assembled or detached		0.40%	3.2	Visual/Manual
106	Trigger peened end over max.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Trigger Pivot				8836974
					NEXT HIGHER ASSEMBLY 8836957, 9288866 or 9313553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Length of small diameter, min.		0.40%	3.2	Gage
Minor 201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.10	Pull Pin		8836960
			NEXT HIGHER ASSEMBLY 8836957, 9288866 or 9313553

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of wire		0.40%	3.2	Gage
102	Crack		0.40%	3.2	Visual
Minor					
201	Length from top loop to end, min.		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2022-10-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Spring, Striker	1 OF 1	8864755
			NEXT HIGHER ASSEMBLY: 8836957, 9288866 or 9313553

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Embrittlement of spring	10	0-1	3.3	4.5.1
102	Torque test	50	0-1	3.2	4.5.8
Minor					
201	Outside diameter		0.65%	3.2	Gage
202	Length over long leg		0.65%	3.2	Gage
203	Length of long leg hook		0.65%	3.2	Gage
204	Diameter of wire		0.65%	3.2	Gage
205	Length over short leg		0.65%	3.2	Gage
206	Length from long leg to short leg		0.65%	3.2	Gage
207	Length over bend of short leg		0.65%	3.2	Gage
208	Length of short leg hook		0.65%	3.2	Gage
209	Overall length		0.65%	3.2	Gage
210	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 20260-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Safety Clip Assembly				8836958
					NEXT HIGHER ASSEMBLY 8836957, 9288866 or 9313553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of wire		0.40%	3.2	Gage
102	Crack		0.40%	3.2	Visual
103	Width of gap		0.40%	3.2	Gage
104	Irregular plating on pull pin		0.40%	3.2	Visual
105	Safety clip assembly spring test	4.4.3.5	4.4.3.5	3.2	4.5.7
Minor					
201	Length from bend to end, min.		0.65%	3.2	Gage
202	Length		0.65%	3.2	Gage
203	Pull pin incorrectly assembled		0.65%	3.2	Visual
204	Width		0.65%	3.2	Gage
205	Spring clip safety missing or mislocated		0.65%	3.2	Visual
206	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.13	Cover		8836965
			NEXT HIGHER ASSEMBLY: 8836957, 9288866 or 9313553

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter of case flange		0.40%	3.2	Gage
102	True position of striker pivot hole		0.40%	3.2	Gage
103	Length from primer hole to striker spring contact shoulder		0.40%	3.2	Gage
104	True position of safety clip holes		0.40%	3.2	Gage
105	Location of lever forward pivot		0.40%	3.2	Gage
106	Width of striker assembly cavity		0.40%	3.2	Gage
107	Thickness of sides of striker assembly cavity		0.40%	3.2	Gage
108	Diameter of primer cavity		0.40%	3.2	Gage
109	Diameter of hinge pin hole(s)		0.40%	3.2	Gage
110	Depth of primer cavity		0.40%	3.2	Gage
111	Depth of intermediate charge cavity		0.40%	3.2	Gage
112	Crack, or evidence of porosity		0.40%	3.2	Visual
113	Length of press fit shoulder		0.40%	3.2	Gage
Minor					
201	Non-reflective finish improper		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026/0-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.14	Spring, Trigger				8836975
					NEXT HIGHER ASSEMBLY 8836973
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Embrittlement of spring	10	0-1	3.3	4.5.1
102	Spring wound improperly		0.40%	3.2	Visual
Minor					
201	Inside diameter of spring		0.65%	3.2	Gage
202	Inside diameter of loop		0.65%	3.2	Gage
203	Angle of free end		0.65%	3.2	Gage
204	Width of spring center to loop center		0.65%	3.2	Gage
205	Length of free leg to spring center		0.65%	3.2	Gage
206	Width of free end		0.65%	3.2	Gage
207	Entire loop not flat		0.65%	3.2	Gage
208	Diameter of wire		0.65%	3.2	Gage
209	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 20260410-19T07:46Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.15	Spool Retainer	1 OF 1	9282921
			NEXT HIGHER ASSEMBLY 9288866

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall width		0.40%	3.2	Gage
102	Width between free ends		0.40%	3.2	Gage
103	Diameter of wire		0.40%	3.2	Gage
104	Free height		0.40%	3.2	Gage
105	Inside length		0.40%	3.2	Gage
106	Spring test	4.4.3.1	4.4.3.10	3.2	4.5.12
Minor					
201	Inside radius		0.65%	3.2	Gage
202	Height of holding leg		0.65%	3.2	Gage
203	Inside angle of free legs		0.65%	3.2	Gage
204	Angle at free end		0.65%	3.2	Gage
205	Length of inside angle		0.65%	3.2	Gage
206	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.16	Wire Spool Assembly				9269197
					NEXT HIGHER ASSEMBLY 8836957, 9288866, 9313553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Nail(s) missing		0.40%	3.2	Visual
102	Wire not pulled through slot		0.40%	3.2	Visual
103	Length of trip wire	20	0-1	3.2	Gage
Minor					
201	Wire above flush		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026Z0-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.17	Cover Loading Assembly	1 OF 1	8836964
			NEXT HIGHER ASSEMBLY: 8836957, 9288866 or 9313553

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component missing		0.40%	3.2	Visual
102	Crimp improper		0.40%	3.2	Visual
103	Disc ruptured, or containing fold		0.40%	3.2	Visual
Minor					
201	Varnish seal missing or incomplete		0.65%	3.2	Visual
202	Varnish where not required		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-Q Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 20264O-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.18	Illuminant Assembly, (Prior To Pressing Of Igniter And Illuminant Composition)	1 OF 1	9269017, 9313554, 9282864
			NEXT HIGHER ASSEMBLY 8836957, 9313553, 9288866

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Interior of case not coated to within 1/4 inch of top with insulating composition		0.40%	3.2	Scale
Minor 201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2022-10-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.19	Illuminant Loading Assembly	1 OF 1	9269017, 9282864, 9313554
			NEXT HIGHER ASSEMBLY: 8836957, 9313553, 9288866

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Depth from top of case to first fire, min.		0.40%	3.2	Gage
102	Inside diameter at mouth (after pressing)		0.40%	3.2	Gage
103	Outside diameter, max. (after pressing)		0.40%	3.2	Gage
104	Ignition composition missing		0.40%	3.2	Gage
105	Top edge of case and interior surface not clean to a depth of 1/8 inch		0.40%	3.2	Visual
106	Case cracked or split		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 20260410-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8836957, 9288866, 9313553
4.4.2.20	Flare Assembly (Prior to Sealing Cover to Body)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Cover Loose		0.40%	3.2	Manual
Minor 201	Evidence of poor workmanship		1.0%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026/10-19T07:46Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.21	Flare Assembly, (Prior To Assembling Lever Or Bracket)		1 OF 1		8836957, 9288866, 9313553
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component missing or inproperly assembled		0.40%	3.2	Visual
102	Seal on cover or primer missing or incomplete		0.40%	3.2	Visual
103	Sealing compound where not required		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 20260-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.22	Flare Assembly	1 OF 2	8836957, 9288866, 9313553
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Width to which bracket is tightened, max.		100%	3.2	Gage
2	Safety clip missing or incorrectly attached		100%	3.2	Visual
Major					
101	Length to bracket from base of flare		0.40%	3.2	Gage
102	Any component missing or damaged		0.40%	3.2	Visual
103	Striker restraining pin not removed		0.40%	3.2	Visual
104	Spool of wire missing or not secure (alt. only)		0.40%	3.2	Visual/Manual
105	Jumble Test	4.4.3.1	4.4.3.2	3.4	4.5.2
106	Functioning (Ambient Temp.)		4.4.3.3.1	3.5.1	4.5.4.1
107	Functionig (Cold Temp.)		4.4.3.3.2	3.5.2	4.5.4.2
108	Functioning (Temperature and Humidity Cycling)		4.4.3.3.3	3.5.3	4.5.4.3
109	Candlepower		4.4.3.4	3.2	4.5.6

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 20260-19T07:46Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.22	Flare Assembly	2 OF 2	8836957, 9288866, 9313553
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Safety clip under bracket prongs		0.65%	3.2	Visual
202	Marking misleading or unidentifiable		0.65%	3.2	Visual
203	Pin pull inside trigger assembly		0.65%	3.2	Visual
204	Paint on areas where not permitted		0.65%	3.2	Visual
205	Width to which bracket is tightened, min.		0.65%	3.2	Gage
206	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.23	Box, Small, (Unsealed)				8830881
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Incorrect number of spools of wire		0.40%	3.2	Visual
Minor 201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2020640-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.24	Box, Large, (Unsealed)	1 OF 1	8830881
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Small carton missing		0.40%	3.2	Visual
102	Flare assembly missing		0.40%	3.2	Visual
103	Instruction sheet missing		0.40%	3.2	Visual
Minor					
201	Support missing		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-(Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.25	Box, Large				8830881
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Carton flap not secured		0.40%	3.2	Manual
102	Box damaged to the extent that the contents are exposed or likely to become exposed in subsequent handling		0.40%	3.2	Visual
Minor					
201	Contents move when shaken		0.65%	3.2	Manual
202	Marking misleading or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current revision before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.26	Bag				8830881, 9277238
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bag seal incomplete or incorrectly located		0.40%	3.2	Visual
102	Bag, cut, torn or punctured		0.40%	3.2	Visual
103	Water submersion	10	0-1	3.6	4.5.5
Minor					
201	Marking misleading or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.27	Box, Packing				8830880, 9277239
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Top improperly assembled		0.40%	3.2	Visual/Manual
102	Board broken or split		0.40%	3.2	Visual/Manual
103	Strapping missing, broken or loose		0.40%	3.2	Visual/Manual
104	Box damaged to the extent that contents are exposed or likely to become exposed in subsequent handling		0.40%	3.2	Visual
Minor					
201	Strapping improperly assembled		0.65%	3.2	Visual/Manual
202	Contents loose when shaken		0.65%	3.2	Manual
203	Metallic seal missing, unsealed or improperly positioned		0.65%	3.2	Visual
204	ICC marking incorrect, incomplete or illegible		0.65%	3.2	Visual
205	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assistdocs.com - Downloaded: 2026-01-19T07:40Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.28	Box, Fiberboard, Packing (Alt.) (Prior To Sealing)				9277238
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Flare assembly missing		0.40%	3.2	Visual
102	Spool of wire and nails missing or insecure		0.40%	3.2	Visual/Manual
103	Any packing component missing		0.40%	3.2	Visual
104	Instruction sheet missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 20260-19T07:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-20420J(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.29	Box, Packing (Alt.)		9277238
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outer flaps of box not secured		0.40%	3.2	Manual
102	Label missing		0.40%	3.2	Visual
103	Box damaged to extent that contents are exposed or liable to become exposed		0.40%	3.2	Visual
Minor					
201	Contents loose when shaken		0.65%	3.2	Manual
202	Marking misleading or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 20260210-190707462
Check the source to verify that this is the current version before use.

4.4.3 Testing. The following tests shall be performed using MIL-STD-105 for sampling plan requirements unless otherwise specified.

4.4.3.1 Jumble test. (See 3.4). The flare assemblies shall be observed and examined visually without disassembly for evidence of failure to comply with the requirements as classified in Table II.

TABLE II - Jumble Test Requirement

Defect	Classification
Flare functions during test (see 3.4.1)	Critical
Safety clip assembly totally or partially disengages during test (see 3.4.2)	Critical
Flare assembly not safe to transport after test (see 3.4.3)	Critical
Flare assembly damaged after test (see 3.4.4)	Major

4.4.3.1.1 First three lots - Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the requirements fifty (50) flare assemblies shall be selected from each lot for this test. The lot shall be rejected if during the test a critical defect occurs or four (4) or more assemblies exhibit the major defect listed in Table II. The test procedure shall be as specified in 4.5.2.

4.4. 3.1.2 Reduced sampling after three lots - After three (3) consecutive lots have compiled with the criteria of 4.4.3.1.1, twenty (20) flare assemblies shall be selected from each lot for test l The lot shall be rejected if a critical defect occurs or two (2) or more assemblies exhibit the major defect listed in Table II. The test procedure shall be as specified in 4.5.2.

4.4.2.2 Moisture or volatile content as applicable (see Table Major defects.

TABLE III - Moisture or Volatile Content

Material	See Dwg. (when applicable)
Ignition composition	(9269017 or 9313554)
Baffle (paper)	(8836957 or 9313553)
Intermediate Charge	(8836964 or 9288873)
Illuminant composition	(9269017 or 9313554)
First fire composition	(9269026 or 9282865)
Asbestos	(9269018 and 9269105)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

The contractor shall provide adequate controls to insure compliance with the requirements and shall select a three to four ounce sample of each material, listed in Table III, from each eight hours of production and place it in an air tight labeled container indicating the name of the material and the date of production. If the sample, after analysis, fails to comply with the applicable requirements, the eight hours of production represented by the sample shall be rejected. Test procedures shall be as specified in 4.4.3.

4.4.3.3 Functioning. The functioning test consists of three (3) distinct phases: Ambient, Cold, and Temperature-Humidy Cycling. Reduced sampling will be authorized only when three (3) consecutive lots have met the acceptance criteria specified in all three phases. Compliance with the criteria in any one phase will not result in reduced sampling for that phase. The flare assemblies shall be observed for any evidence of failure to comply with the requirements as classified in Table IV when tested as specified in 4.5.4.

TABLE IV - Functioning Requirement

Defect	Classification
Flare explodes (see 3.5)	Critical
Flare burns less than 45 secs. or fails to ignite (see 3.5)	Major
Burning time less than 55 secs. but not less then 45 secs. (See dwg. 8836957 or 9288866 when applicable)	Major
Trigger and pin pull force incorrect (see 3.5.1 and 3.5.3) but pin pull force not less than 1.00 pound	Major
Pull pin force less than 1.00 pound (see 3.5.1 and 3.5.3)	Major

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

4.4. 3.3.1 Ambient temperature. (See 3.5.1).

4.4.3.3.1.1 First three lots. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified, eighty (80) flare assemblies shall be selected from each lot for this test. The lot shall be rejected if:

a. A critical defect occurs.

b. A total of three (3) or more flares burn less than 45 secs. or fail to ignite.

c. Any pull pin force is less than 1.00 pound.

d. A total of four (4) or more flares exhibit any of the remaining major defects listed in Table IV.

4.4.3.3.1.2 Reduced sampling after three lots. After three (3) consecutive lots have compiled with the criteria of 4.4.3.3.1.1, thirty-two (32) flare assemblies shall be selected from each lot for test. The lot shall be rejected if:

a. A critical defect occurs.

b. A total of two (2) or more flares burn less than 45 secs. or fail to ignite.

c. Any pull pin force is less than 1.00 pound.

d. A total of three (3) or more flares exhibit any of the remaining major defects listed in Table IV.

4.4.3.3.2 Cold temperature. (See 3.5.2).

4.4.3.3.2.1 First three lots. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified, eighty (80) flare assemblies shall be slected from each lot for this test. The lot shall be rejected if:

a. A critical defect occurs.

b. A total of three (3) or more flares burn less than 45 secs. or fail to ignite.

c. A total of five (5) or more flares exhibit any of the remaining major defects listed in Table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

4.4.3.3.2.2 Reduced sampling after three lots. After three (3) consecutive lots have complied with the criteria of 4.4.3.3.2.1, thirty-two (32) flare assemblies shall be. selected from each lot for this test. The lot shall be rejected if:

a. A critical defect occurs.

b. A total of two (2) or more flares burn less than 45 secs. or fail to ignite.

c. A total of four (4) or more flares exhibit any of the remaining major defects listed in Table IV.

4.4.3.3.3 Temperature and humidity cycling. (See 3.5.3).

4.4.3.3.3.1 First three lots. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified eighty (80) flare assemblies shall be selected from each lot for this test. The lot shall be rejected if:

a. A critical defect occurs.

b. A total of three (3) or more flares burn less than 45 secs. or fail to ignite.

c. Any pull pin force is less than 1.00 pound.

d. A total of four (4) or more flares exhibit any of the remaining major defects listed in Table IV.

4.4.3.3.3.2 Reduced sampling after three lots. After three (3) consecutive lots have complied with the criteria of 4.4.3.3.3.1, thirty-two (32) flare assemblies shall be selected from each lot for this test. The lot shall be rejected if:

a. A critical defect occurs.

b. A total of two (2) or more flares burn less than 45 secs. or fail to ignite.

c. Any pull pin force is less than 1.00 pound.

d. A total of three (3) or more flares exhibit any of the remaining major defects listed in Table IV.

4.4.3.4 Candlepower. (See dwg. 8836957, 9313553 or 9288866, when applicable) - Major defect.

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:46Z
Check the source to verify that this is the current version before use.

4.4.3.4.1 First three lots. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the requirements , eighty (80) flare assemblies shall be selected from each lot for this test. The lot shall be rejected if, during the test, three (3) or more flare assemblies fail to comply with the requirement. The test procedure shall be as specified in 4.5.6.

4.4.3.4.2 Reduced sampling after three lots. After three (3) consecutive lots have compiled with the criteria of 4.4.3.4.1, thirty-two (32) flare assemblies shall be selected from each lot for this test. The lot shall be rejected if, during the test, two (2) or more flare assemblies fail to comply with the requirement. The test procedure shall be as specified in 4.5.6.

4.4.3.5 Safety clip assembly spring test. (See dwg. 8836958) - Major defect.

4.4,3.5.1 First three lots. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the requirements, eighty (80) safety clip assemblies shall be selected from each lot for this test. The lot shall be rejected if, during the test, three (3) or more safety clip assemblies fail to comply with the requirement. The test procedure s-nail be as specified in 4.5.7.

4.4.3.5.2 Reduced sampling after three lots. After three (3) consecutive lots have complied with the criteria of 4.4.3.5.1, thirty-two (32) safety clip assemblies shall be selected from each lot for this test. The lot shall be rejected if, during the test, two (2) or more safety clip assemblies fail to comply with the requirement. The test procedure shall be as specified in 4.5.7.

4.4.3.6 Determination of composition. Determination of composition or batch record, as applicable (see Table V).

TABLE V - Determination of Composition

Defect	Classification
Illuminant composition (see 9269024 or 9313565 as applicable)	Major
Intermediate charge (batch record) (see 8836964 or 9288873 when applicable)	
First fire composition (batch record) (see 9269026 or 9282865, when applicable)	
Ignition composition (see 9269025 or 9313566 when applicable)	Major

Source: https://assist.dla.mil -- Downloaded: 20260-19T07:46Z
Check the source to verify that this is the current version before use.

The contractor shall provide adequate controls in insure compliance with the requirements and shall select a three to four ounce sample of the illuminant and ignition compositions from each eight hours of production and place it in an air tight labeled container indicating the name of the material and the date of production. If the sample after analysis fails to comply with the applicable requirements, the eight hours of production represented by the sample shall be rejected.

4.4.3.7 Protective coating test of clip, safety assembly (dwg. 8836958), lever (dwg. 8836961), Cover (dwg. 8836965), trigger (8836976), striker assembly (dwg. 8836971), pivot, trigger (dwg. 8836976), spring, trigger (dwg. 8864755), trigger (dwg. 8836976), washer trigger (dwg. 8836977), spring (dwg. 8864755) and spool retainer - Major defects. Sampling, acceptance and rejection shall be in accordance with specification QQ-P-416 except that actual parts shall be used in lieu of specimens. The test procedure shall be as specified in 4.5.10.

4.4.2.8 Check test for possible deterioration of percussion primer. (See MIL-P-20444). If the total elapsed time between original acceptance of any primer lot and the assembly of that let into the flare assemblies exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall be subjected to and must satisfactorily comply with the check test for deterioration specified in MIL-P-20444. The check test shall be performed by the contractor (see 6.5) prior to assembling the primers into the flares.

4.4.3.9 Spool retainer spring test. (See dwg. 9282921) - Major defect.

4.4.3.9.1 First three lots. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the requirement , eighty (80) spool retainers shall be selected from each lot for this test. The lot shall be rejected if, during the test, three (3) or more spool retainers fail to comply with the requirement. The test shall be as specified in 4.5.12.

4.4.3.9.2 Reduced sampling after three lots. After three (3) consecutive lots have complied with the requirements of 4.4.3.10.1, thirty- two (32) spool retainers shall be selected from each lot for this test. The lot shall be rejected if, during the test, two (2) or more spool retainers fail to comply with the requirements. The test shall be specified in 4.5.12.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the table starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of tune contract. See Section 6 of MIL-A-48078 and 6.- herein.

4.5 Test methods and procedures

4.5.1 Embrittlement of striker spring, trigger spring and retainer. The spring shall be stretched along its longitudinal axis until it no longer returned to its original length or profile it shall be considered permanently distorted. Any spring that breaks during the test shall be classed defective. Springs subjected to test shall not be returned to the lot. (Destructive Test)

4.5.2 Jumble test. The flare assemblies shall be tested in accordance with the procedure specified in MIL-STD-331 except that the test duration shall be twenty (20) plus or minus 1 minute. (Destructive Test)

4,5.3 Moisture content or volatile content, as applicable. (Destructive Test)

4.5.3.1 Baffle and asbestos. The moisture content shall be determined in accordance with Method 102.1 of MIL-STD-1234.

4.5.3.2 Intermediate charge composition. The volatile content of the intermediate charge composition shall be determined in accordance with 4.5.3.1.

4.5.3.3 Illuminant and ignition composition. The moisture content of the illuminant and ignition composition shall be determined as follows: A 10 gram sample shall be added to a flask with 100 ml. of methyl alcohol. The flask shall be covered and allowed to stand for one-half hour. A 50 ml. aliquot shall be removed from the flask and added to the Karl Fischer flask. The sample shall be titrated with the Karl Fischer reagent. A blank with methyl alcohol shall be titrated in the same manner as above. The Karl Fischer reagent shall be standardized as specified in MIL-STD-1234, Method 101.2 except F shall be mg/ml. The calculation for moisture content shall be as follows:

$$\text{Percent water } (H_2O) = \frac{(V-S)\ F}{5 \times 10}$$

Where:

V = Ml. of KF used to titrate sample moisture.
F = Standardization factor.
S = Ml. of KF used to titrate blank moisture.
10 = Factor to convert to percent.
5 = Weight of sample contained in aliquot used.

4.5.3.4 First fire composition. The volatile content of the sample shall be determined as specified in 4.5.3.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

4.5.4 Functioning

4.5.4.1 Ambient temperature (21 ± 2.8°C) (70 ± 5°F). The flares shall be conditioned for a min. of 16 hours at ambient temperature. After conditioning, the flares shall be functioned within 5 minutes utilizing the test equipment specified on dwg. 9247118, using the operating instructions specified on dwg. 9247119. The flare assemblies shall be fired, the pull forces recorded and burning times of all flares shall be determined by means of an approved measuring device. (Destructive Test)

a. One-half of the sample flares shall be set-up with the trip wire connected to the pull pin that has been assembled in lieu of the safety clip. The flares shall be fired and observation made for compliance with the applicable requirements.

b. One-half of the sample flares shall be set-up with the trip wire connected to the trigger. One-half of these flares shall be fired as specified in “a” and observation made for compliance witht he applicable requirement. The remaining half of the flares shall be fired by cutting the trip wire and observation made for compliance with the applicable requirement.

4.5.4.2 Cold temperature (minus 34.5 ± 2.8°C) (minus 30 ± 5°F). The flares shall be conditioned for a min. of 16 hours at a temperature of (minus 34.5 ± 2.8°C) (minus 30 ± 5°F). The flares shall be taken from the conditioning chamber and fired within 5 minutes by removing the pull pin. The burning time and the pull pin force shall be determined by the same method used in 4.5.4.1. (Destructive Test)

4. 5.4.3 Temperature and humidity cycling. The flare assemblies shall be tested in accordance with the procedure specified in MIL-STD-331, except that the time duration of the test shall be limited to the first 48 hours, and timing shall not start until the end of the initial conditioning at (minus 54°C) (minus 65 °F). At the end of the test period, the flare assemblies shall be allowed to return to ambient temperature (21 ± 2.8°C) (70 ± 5°F) prior to functioning. The burning time, pin pull force, and trigger pull force shall be determined by the same methods used in 4.5.4.1. (Destructive Test)

4.5.5 Water submersion test. The filled and sealed packages, after conditioning at ambient temperature (21 ± 2.8°C) (70 ± 5°F for at least four (4) hours, shall be tested for leakage by submerging in water in a large drum. The water in the drum shall be maintained at (32.2° ± 2.8°C) (90° ± 5°F). The package shall be held beneath the surface of the water for thirty (30) seconds minimum while observation is made for leakage. The package shall be inverted and the procedure repeated. A steady stream or recurring stream of bubbles from any surface shall be cause for rejection. Where there is no emission of bubbles observed, the package shall be dried, opened and inspected. Evidence of water in the barrier bag shall also be cause for rejection. (Non-Destructive)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

4.5.6 Candlepower test. The flare assemblies shall be supported in a suitable fixture with the ignition end up, and fired by removing the pull pin. The candlepower determination shall be made in accordance with the procedures and equipment specified on dwgs. 9201268, 9201390 and 9201136 using the average candella reading taken during a consecutive burning time period of fifty-five (55) seconds, min. (Destructive Test)

4.5.7 Safety clip assembly spring test. The safety clip assembly shall be placed on the test gage and allowed to remain there for the required time. At the end of the test period, the clip assembly shall be removed from the test gage and the distance between the ends checked for compliance with the dwg. requirement. The equipment used for this test shall be as specified in 4.4.4. (Non-Destructive Test)

4.5.8 Torque test of spring. The springs shall be distorted the required amounts and the distortion forces recorded for compliance with the dwg. requirements. The test equipment used shall be as specified in 4.4.4. Springs subjected to these tests shall not be returned to tune lot. (Destructive Test)

4.5.9 Determination of composition

4.5.9.1 Intermediate charge and first composition. In lieu of making an analysis of each batch of intermediate charge or first fire composition, as applicable, a batch record shall be prepared for each batch produced for production use. This batch record shall be used to demonstrate the actual amounts of each ingredient used to make the batch as specified on the applicable drawing. In addition to recording composition of the batch, the batch record shall also be used to record and control curing time for mixtures as necessary prior to loading. A batch of composition shall be the amount of composition used during a single work shift by a single employee or crew. The batch may contain one or more mixes. When more than one mix is used in a batch, the exact proportion of ingredients shall be measured into each mix.

4.5.9.2 Illuminant and ignition composition. The illuminant and ignition composition shall be analyzed as follows to determine compliance with applicable drawing requirements.

4.5.9.2.1 Magnesium. Atomic Absorption Spectrophotometer Method.

Preparation of standard. Using stock magnesium solution prepare the standard using the following procedure: Weight accurately 1.0 gram of high purity magnesium to the nearest .0001 gram and dissolve in a minimum volume of nitric acid duluted one to one with distilled water. Cool and dilute to one liter. (One milliliter of solution equals 1,000 micrograms of magnesium.)

4.5.9.2.1.2 Procedure. Prepare the sample of material to be tested in the following manner: Weight a one gram sample to the the

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:46Z
Check the source to verify that this is the current version before use.

nearest .0001 gram. A 1.00 gram sample when dissolved and diluted to a liter will have a theoretical concentration of 1,000 micrograms/milliliter. Dissolve in a minimum volume of one to one citric acid. Cool and make proper dilutions to obtain a working range of 20 micrograms/milliliter. Run the analysis according toprocedure outlined in Analytical Methods for Atomic Absorption Spectrophotometer - Perkin - Elmer, Revision March 1971 (or equivalent). The percent of magnesium shall be calculated as follows:

Percent of magnesium = (A X B)/C

where:

A = Reading of the unknown.
B = Theoretical Mg. content of illuminant or ignition composition.
C = Reading of the standard.

Instrument settings:

A. Wave length -285-UV, Slit setting 4; Light source - hollow cathode lamp (use the current indicated on the lamp label or instruction card for Model 403).
B. Oxidant - Air.
C. Fuel - Acetylene.

4. 5.9.2.2 Sodium nitrate. Atomic Absorption Spectrophotometer Method.

4.5.9.2.2.1 Preparation of standard. Prepare a stock standard solution of 1,000 micrograms per milliliter of sodium by dissolving 2.542 grams of sodium chloride in one liter of dionized water.

4.5. 9.2.2.2 Procedure. Using the same master solution from 4.4.9.2.1.1, perform the following operations: Make the proper dilutions to obtain a working range of the standard and sample in the 19-21 micrograms per milliliter range. Since sodium is partially ionized in the air acetylene flame, it is necessary to add another alkali to the standard and sample to nullify the effects of ionization. This may be done by adding 0.2 grams of potassium nitrate to each 100 milliliters of the standard and sample. The anlysis shall be run according to the procedure outlined in Analytical Methods for Atomic Absorption Spectrophotometer - Perkin - Elmer, Revision March 1971 (or equivalent). The percent of sodium shall be calculated as follows:

$$\text{Percent of sodium} = \frac{A \times B}{C}$$

where:

A = Reading of the unknown
B = Theoretical percent of sodium nitrate of mix.
C = Reading of the standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

Instrument settings:
A. Wave length -295- visible, Slit setting 4, Light source - hollow cathode lamp (use the current indicated on the lamp label or instruction card for Model 403)
B. Oxidant - Air
C. Fuel - Acetylene

4.5.9.3 Binder composition. The percentage of binder in the composition shall be calculated by subtracting from 100 the sum of the percentage of sodium nitrate and magnesium.

4.5.9.4 Potassium perchlorate. Transfer an accurately weighed portion of approximately one gram of the sample to a weighed Gooch crucible . Leach with ten 10 ml. portions of hot water and place the crucible, including sample, in a drying oven at 100° to 105°C (212°F to 221°F) for two hours. Cool in a desiccator and weight the crucible with the residue. The percent potassium perchlorate shall be calculated as follows:

$$\text{Percent of Potassium Perchlorate} = \frac{100.0\ (A)}{W}$$

where:
A = Loss in weight of the sample.
W = Weight of sample on a moisture free basis.

4.5.9.5 Batch records (alternate method). In lieu of making an analysis of each batch of illuminant and ignition composition, as applicable, a batch record shall be prepared for each batch produced for production. Batch records shall be prepared as specified in 4.5.9.1

4.5.10 Protective coating of applicable parts. (See 4.4.3.10). The adequacy of the protective coating shall be tested in accordance with the procedures specified in Specification QQ-P-416.

4.5.11 Length of trip wire. The trip wire shall be removed from the spool assembly and the length of the wire measured to determine compliance with the requirement specified. (Non-Destructive Test)

4.5.12 Spool retainer spring test. The spool retainer shall be placed on the test gage and allowed to remain there for the required time. At the end of the test period, the spool retainer shall be removed from the test gage and the free height checked for compliance with the dwg. requirement. The equipment used for this test shall be as specified in 4.4.4. (Non-Destructive Test)

5. PACKAGING

5.1 Preservation and packaging

5.1.1 Level A. The flares shall be preserved and packaged in accordance with dwg. 8830881 or 9277238 when applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

5.2 Packing

5.2.1 Level A. The boxes shall (from 5.1.1) be packed in accordance with dwg. 8830880 or 9277239, when applicable.

5.3 Marking. The boxes shall be marked as specified on dwgs. 8830881 and 8830880, or 9277239 and 9277238, when applicable.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on the Flare M49A1.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAT-I, Dover, NJ 07801,

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Cost of check test. The Contracting Officer should arrange for the contractor to be reimbursed for the expense incurred in the performance of the check tests. The tests shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the primers in the flares and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage , etc. permitted while the primer lots were under the jurisdiction of either contractor.

6.6 Safe transport - The flare assembly shall be considered safe to transport provided no evidence exists of totally or partially disengaed safety clips, loose powder or composition in the box, or missing, loose, or protruding primers.

6.7 Flare assembly - The flare assembly will be considered free of damage that will affect the intended functioning provided the cover seal has not been broken by movement, or the trigger has not been broken off or disengaged from the bracket.

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:46Z
Check the source to verify that this is the current version before use.

6.8 Proving ground functioning test summary:

Test	Production Lots			Requirements
	First Article	First Three Consecutive Acceptable	Remaining	
Functioning	128	240	96	See 3.5 and Table IV

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1370-A131)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

COMMANDER
US ARMY ARMAMENT RESEARCH & DEVELOPMENT COMMAND
DRDAR-QAT-I
DOVER, NJ 07801

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-20420J	Flare, Surface, Trip, M49A1 Parts, and Loading, Assembling & Pack.

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-01-19T07:46Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:46Z
Check the source to verify that this is the current version before use.